



Carrier network for AI Inference

Side Meeting on IDN - Inference Distribution Network Architecture
and Technologies

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AI Inference as Carrier Network Problem

- AI inference is moving from centralized cloud execution toward distributed, latency-sensitive, context-dependent service delivery
- Requests may be interactive, bursty, geographically distributed and privacy-sensitive
- Aspects to take into consideration:
 - Latency (incl. tail latency)
 - Impact on backbone traffic
 - Dependency on DC locations where inference is running (could be from different administrative domains)

Inference Distribution Network

- CDNs distribute content
- AI inference has different distribution considerations
 - Execution, context, model-related state, knowledge availability, etc
- For inference, the best endpoint is not necessarily the nearest server, but the endpoint from where the inference can be best served
 - It will be the node with the right combination of compute, network conditions, context and inference state

The role of the Carrier Network

- The role of the carrier network could be to expose capabilities to select, reach and use the most appropriate inference execution point
 - Inference-aware reachability (e.g., model and service availability)
 - Compute-aware steering (CATS related)
 - Context-aware routing (context availability or freshness)
 - Policy-aware delivery (e.g., sovereignty)
 - Others (e.g., energy-efficiency)

Potential items for further development in IETF

- Inference service discovery
- Inference-aware steering
- Context handling / advertisement
- Multidomain inference (cross-domain inference peering)